CHISOMO MBEWE

STUDENT

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KUALA LUMPUR, MALAYSIA

PROFILE

I am a final-year Electrical and Electronic Engineering student at Asia Pacific University of Technology & Innovation with hands-on experience across embedded systems, IoT, power system analysis and AI-driven software. I enjoy taking projects from concept to working prototype, from ESP32 firmware and sensor integration to live dashboards and power system studies in ETAP and MATLAB, and I work well both independently and as part of a team.

EXPERIENCE

2023 – Present

Student

Asia Pacific University of Technology & Innovation

- Developed the firmware, sensor calibration and live web dashboard for W.A.V.E, a solar-powered boat that monitors water quality and uses on-board AI to classify it. Our team of five won Grand Champion at AgriXcelerate 2026. [2026]
- Programmed a dual-core ESP32-S3 (FreeRTOS) to read an RS-485 Modbus water sensor and send live data over 4G to Firebase, displayed on a Next.js dashboard with a GPS map. [2026]
- Represented APU at EPSSCOM 2026, a national power system simulation competition organized by IEEE PES Malaysia. Responsible for the protection coordination study of a 132/11 kV industrial network with distributed energy resources, modelled in ETAP. [2026]
- Developed "RecipeRun," an AI-powered cross-platform cooking assistant mobile application using Flutter, Firebase, TypeScript, and a hybrid AI routing pipeline (Google Gemini and OpenAI). [2026]
- Engineered "TrustPort," an AI-driven fintech and credit scoring web platform utilizing Python, FastAPI, and a multi-model AI ensemble for visual fraud detection and financial reasoning. [2025]
- Designed a simulation of an energy harvesting circuit capable of harvesting electrical energy from piezoelectric devices using MATLAB Simulink. [2025]

- Designed and implemented an audio enhancement application in MATLAB utilizing four distinct enhancement techniques/filters to address various noise types and improve signal clarity. [2025]
- Developed a web application using Streamlit that makes use of a pretrained AI model that detects a person's diabetic status. [2025]
- Designed a Demand Side Management Scheme in MATLAB Fuzzy Logic Designer to optimize energy consumption, as part of a school assignment. [2025]
- Developed a smart energy consumption meter using MATLAB App Designer as part of a school assignment. [2024]
- Designed and simulated a hydroelectric power station model in LABVIEW, showcasing control simulation skills as part of a school assignment. [2024]
- Designed lighting schemes for four rooms using Relux and created the room schematics in AutoCAD as part of a university assignment. [2024]

May 2026 – Feb 2027 **Part-time Student Intern**
Asia Pacific University of Technology & Innovation, AGRHI HUB

- Continued developing and maintaining the Autopot monitoring and fertigation IoT systems built during my student internship. [2026 – 2027]

2025 – 2026 **Electrical and Electronic Engineering Student Intern**
Asia Pacific University of Technology & Innovation, AGRHI HUB

- Engineered a dual-system IoT solution for monitoring and fertigation using ESP32 and ADS1115. [2025 – 2026]
- Built a cross-platform Flutter dashboard with Firebase for real-time pH, EC, and moisture tracking. [2025 – 2026]
- Developed logic for automated nutrient and water delivery via scheduling and manual override. [2026]
- Trial-tested the system over 30 days and authored a research paper on it, "IoT Design for Automated Plant Monitoring and Fertigation Management". [2026]

2023 – 2025 **School Project Group Leader**
Asia Pacific University of Technology & Innovation

- Developed an application integrating four distinct machine vision AI models, which were each built from scratch by team members, to identify medical conditions such as benign breast tumors, retinal fundus diseases, COVID-19, and MRI-based brain age estimation. [2025]
- Worked closely with group members to create a smart hospital system which incorporates various electronics and

microcontrollers, including Arduinos and ESP32s, enabling real-time monitoring. [2025]

- Collaborated on the ideation of “DripTech”, a smart automated irrigation system utilizing advanced sensors and a mobile application for remote crop monitoring. [2024 – 2025]
- Guided group members in designing an Arithmetic Logic Unit (ALU); responsible for compiling the final design and preparing the simulation in NI Multisim. [2024]
- Led team members in designing an amplifier circuit. [2024]
- Led a group project to develop a student semester planner application using C-programming. [2023]
- Led a group project to design a 3D model of a robot in SOLIDWORKS. [2023]

2025 **Document Editor**
SureTrack Logistics

- Enhanced document quality by meticulously proofreading and editing content for clarity, grammar, and punctuation.
- Edited and revised a wide array of content for accuracy, clarity, and consistency.
- Edited and reviewed content for grammar, spelling, and punctuation.

2025 **Proofreader**
SureTrack Logistics

- Checked and corrected grammatical and spelling errors in diverse documents.
- Demonstrated strong attention to detail when working with complex technical documents containing specialized terminology or jargon.
- Facilitated timely project completion with efficient task prioritization and time management skills.
- Ensured consistency in style and tone by adhering to clients' specific guidelines and preferences.

SKILLS & ABILITIES

- Strong commitment to continuous improvement
- Effective in both group and individual instruction
- Excellent time management abilities

- Accurate and efficient data entry
- Proficient in English
- Strong reading comprehension and proofreading skills
- Experienced in Python, Dart, HTML, React and C programming in conjunction with Claude Code and Codex
- Experienced in website, mobile and web app development using Flutter, FastAPI and React
- Proficient in integrating advanced AI APIs (OpenAI, Google Gemini) and Cloud backend services (Firebase)
- Skilled in IoT hardware orchestration using microcontrollers (ESP32, Arduino) and sensors
- Skilled in using SOLIDWORKS for 3D modeling and design
- Proficient in MATLAB for simulation and analysis
- Experience creating lighting schemes using Relux
- Skilled in translating AI and human instructions into actionable and tangible results

EDUCATION

March 2023 – Present

Bachelor's of Electrical and Electronic Engineering
Asia Pacific University of Technology & Innovation

- Completed Coursework: Machine Vision and Intelligence, 2025
- Completed Coursework: Digital Signal Processing, 2025
- Completed Coursework: Power Electronics and Drives, 2025
- Completed Coursework: Microprocessor Systems and Embedded Software, 2025
- Completed Coursework: Engineering Project Management, 2025
- Completed Coursework: Digital Electronics, 2024
- Completed Coursework: Analogue Electronics, 2024
- Completed Coursework: Electrical Machines, 2024
- Completed Coursework: Signals and Linear Systems, 2024
- Completed Coursework: Electromagnetic Field Theory, 2024
- Completed Coursework: Python & C-Programming, 2023

April 2025 – Sept 2025

Rocheston Certified AI Engineer Program
Asia Pacific University of Technology & Innovation

- Completed Coursework: Rocheston AI Engineering Course
- Completed Coursework: Microsoft Azure AI Fundamentals

REFERENCES

NICHOLAS BANDA – HEAD of HUMAN RESOURCES, RUBIS ENERGY ZAMBIA LTD

Email Address: Nicholas.Banda@rubiszambia.com

TS. SURESH GOBEE – SENIOR LECTURER, ASIA PACIFIC UNIVERSITY OF TECHNOLOGY & INNOVATION | HEAD of ASIA PACIFIC CENTER OF ROBOTICS ENGINEERING (APCORE)

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